

Explorer

C Quick_so...



Submit

Debugger

```
1 #include <stdio.h>
2
3 void quickSort(int arr[], int low, int high) {
4     if (low >= high) return;
```

Average time

0.467 s

466.75 ms



Maximum time

0.918 s

918.00 ms



✓ 2 out of 2 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 73 ms

Debug



Expected output

4

1 8 3 4

1·8·3·4·

1·3·4·8·

Actual output

4

1 8 3 4

1·8·3·4·

1·3·4·8·

✓ Test case 2 716 ms

Debug



Expected output

7

2 9 -7 0 2 -4 1

2·9·-7·0·2·-4·1·

-7·-4·0·1·2·2·9·

Actual output

7

2 9 -7 0 2 -4 1

2·9·-7·0·2·-4·1·

-7·-4·0·1·2·2·9·

Terminal

Test cases

```
#include <stdio.h>
```

```
void quickSort(int arr[], int low, int high) {  
    if (low >= high) return;  
    int pivot = arr[high];  
    int i = low;  
    int j = high - 1;  
    int temp;  
    while (i <= j) {  
        while (i <= j && arr[i] < pivot) {  
            i++;  
        }  
        while (i <= j && arr[j] > pivot) {  
            j--;  
        }  
        if (i <= j) {  
            temp = arr[i];  
            arr[i] = arr[j];  
            arr[j] = temp;  
            i++;  
            j--;  
        }  
    }  
    temp = arr[i];  
    arr[i] = arr[high];  
    arr[high] = temp;  
  
    quickSort(arr, low, i - 1);  
}
```

```
        quickSort(arr, i + 1, high);

    }

    void printArray(int arr[], int size) {
        for (int i = 0; i < size; i++) {
            printf("%d ", arr[i]);
        }
        printf("\n");
    }

    int main() {
        int n;
        // Input the size of the array
        scanf("%d", &n);
        int arr[n];

        // Input the elements of the array
        for (int i = 0; i < n; i++) {
            scanf("%d", &arr[i]);
        }

        // Print the original array
        printArray(arr, n);

        // Sort the array using quick sort
        quickSort(arr, 0, n - 1);

        // Print the sorted array
        printArray(arr, n);

        return 0;
    }
```